Course Project Proposal

Project Type: Software Development

Process Change Impact Simulator

I plan to develop a Python application that helps organizations understand the possible effects of changing an activity, condition, approval, or rule in an existing business process. My main focus will be on representing the process as a dependency graph and using graph algorithms to trace how a change can move through connected activities. For example, if an approval condition is changed, the system should identify the later tasks, approvals, and outputs that depend on it. I will also include a simulation feature so that users can compare the original process with the changed version and see which parts may be affected.

The system will allow users to create or import a process containing activities, decision points, dependencies, inputs, and outputs. Each activity will have information such as its name, type, required conditions, and connections to other activities. I will design the application to support different types of dependencies, including direct activity dependencies, approval dependencies, and rule-based dependencies. When a user selects a component to change, the system will use graph traversal algorithms to find its downstream dependencies and classify the possible impact. I will also implement a simulation engine that follows the process before and after the change, allowing users to compare affected paths, skipped activities, newly required steps, and changes in final outputs. The application will provide an interactive Streamlit interface for creating process models, running impact analysis, viewing the dependency graph, and reviewing simulation results.

I expect this system to help organizations assess process changes before applying them to real operations. It can reduce the chance of missing dependent activities, improve the review of workflow changes, and make the effects of a proposed change easier to understand. Instead of

checking every connected task manually, users will be able to trace the impact from one selected component and see the possible downstream consequences. The comparison between the original and modified process can also help users identify unexpected changes in process behavior.

For the implementation, I will use Python, Streamlit for the application interface, NetworkX for graph representation and graph algorithms, Pandas for handling process and dependency data, NumPy for numerical calculations used in simulation, and Matplotlib and Plotly for visualizing dependency structures and comparing simulation results. I may use PyYAML or JSON for importing and exporting process definitions. I will create sample process models based on realistic organizational workflows and test the system with different dependency structures, including branching processes, multiple approval paths, circular dependencies, and changes that affect several downstream activities. The project will include the design of the dependency model, implementation of the analysis and simulation algorithms, interface development, testing, and evaluation of the results.

I have used Google Gemini LLM

My Prompt:

I am considering a Python project that can help an organization understand the impact of changing one step, approval, condition, or rule in an existing business process. I want to represent the process using connected activities and dependencies. Help me define the main functions this type of application should have and how it could identify downstream effects of a proposed change.

Gemini Response:

The application could represent each process activity as a node and the relationships between activities as dependencies. A user could select one activity, approval, condition, or rule and

analyze the components that depend on it. The system could trace downstream paths, identify affected activities, and show whether a change could cause steps to be skipped, added, or modified. A comparison between the original process and the changed process could help users understand the possible impact before applying the change.

My Prompt:

For this Process Change Impact Simulator, I need to demonstrate the application without using confidential organizational information. What type of sample process data should I create, and what different scenarios should I test to show that the dependency analysis and simulation are working?

Gemini Response:

I could create sample process data containing activities, decision points, approvals, conditions, dependencies, inputs, and outputs. I could use realistic workflows such as employee onboarding, purchase approvals, or expense approvals. The test data could include normal sequential processes, branching paths, multiple approval paths, and dependencies where one change affects several later activities. I could also create cases with circular dependencies and compare the original process with a modified version to show which activities and outputs are affected. This would allow me to demonstrate the system using data created specifically for the project without using confidential business information.

Data Available for System Demonstration

For the demonstration of my system, I will create synthetic process data using publicly documented organizational workflows as references rather than using arbitrary or randomly created processes. I will use the Business Process Model and Notation (BPMN) standard from the Object Management Group to define the basic structure of the sample processes, including activities, sequence flows, and gateways used for branching and merging. For

specific workflow examples, I will use publicly available documentation from IBM and Oracle. IBM documentation provides examples of onboarding workflows involving activities such as completing required information, granting access, testing a connection, and transitioning to support, as well as approval workflows with multiple approval steps. Oracle's public procurement documentation provides concrete examples of purchase-order approval rules, including conditions based on purchase amount, category, requester, cost center, and contract terms, with approvals that can occur in sequence or in parallel. I will use these documented workflow characteristics to manually create simplified synthetic versions of processes such as employee onboarding and purchase approval in JSON or CSV format. Each process record will contain activities, decision conditions, approval requirements, dependencies, inputs, and outputs. I will then create controlled variations of the same process, such as changing an approval threshold, removing an approval condition, or changing a dependency, and compare the original and modified versions to evaluate whether the simulator correctly identifies the downstream activities and outputs affected by the change. The synthetic datasets will be created specifically for this project and will not contain confidential organizational information.